

Generation Interconnection System Impact Study Report

For Queue Project Q14-0383

Flatlick 500 kV
September 16, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0383. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Flatlick 500 kV substation in the Ohio area. The request was received on September 21, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 170 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$20,881,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GAVIN POWER LLC 500-345 kV Auto	FLATLICK – INTERNATIONAL PAPER 500 kV	100.5	1,310	36.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GAVIN POWER LLC 500-345 kV Auto	Replace transformer with higher-rated unit	\$25,323,000	82% (\$20,881,000)
	Total allocated to project		\$20,881,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.